Recurrence / Transience

Prop^n:- If i is rec and $i \leadsto j$ then j must be rec.

Pf:- $\exists$ some n_0 s.t. $p_{ij}^{(n_0)} > 0$.

If $j \not\leadsto i$, $\Rightarrow p_{ji}^{(n)} = 0 \;\forall\, n \geq 0$.

$= P(X_n = i \mid X_0 = j)$.

Since i is rec,

$$P(X_n = i \text{ for some } n > n_0 \mid X_0 = i) = 1.$$

$$= \sum_{k \in S} P(X_n = i \text{ for some } n > n_0,\ X_{n_0} = k \mid X_0 = i)$$

$$p_{ik}^{(n_0)} P(X_n = i \text{ for some } n \geq 1 \mid X_0 = k) \leq p_{ik}^{(n_0)}$$

$$= \sum_{k \neq j} + P(X_n = i \text{ for some } n > n_0 \mid X_{n_0} = j, \cancel{X_0 = i}) \to 0.$$

$$P(X_{n_0} = j \mid X_0 = i) = p_{ij}^{(n)}$$

$$P(X_n = i \text{ for some } n \geq 1 \mid X_0 = j) = 0.$$

$$\leq \sum_{k \neq j} p_{ik}^{(n_0)} = 1 - p_{ij}^{(n_0)} < 1$$

$$P = \begin{array}{c|cccccc} & 0 & 1 & 2 & \cdots & & \\ \hline 0 & q & p & 0 & & & \\ 1 & q & 0 & p & & & \\ 2 & & q & 0 & p & & \\ & & & & \ddots & & \end{array}$$

$0 < p < 1$

$p_{00} = q > 0$

'0' has periodicity 1, '0' aperiodic.

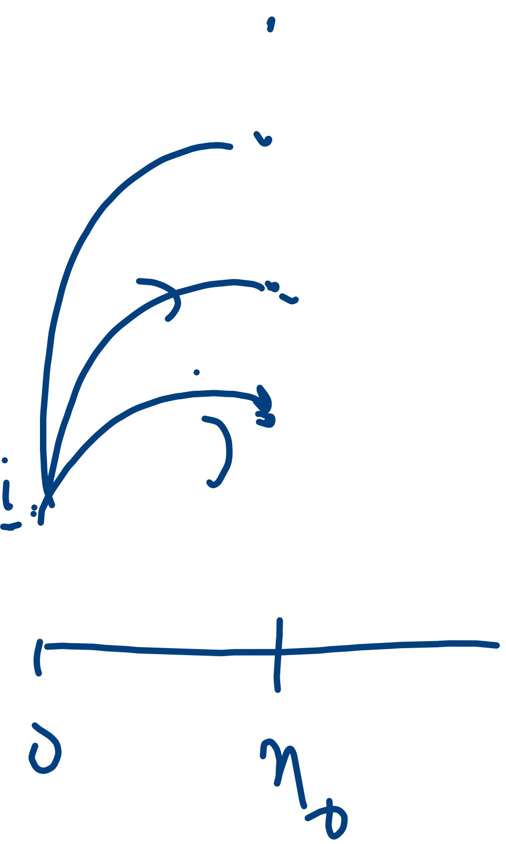
The chain is irr.

$p_{01}^{(1)} > 0$, '0' $\rightsquigarrow$ 1 and $1 \rightsquigarrow 0$.

$p_{10}^{(1)} > 0$

If $p > q$, 0 is tr.

$p \leq q$ 0 is rec.



$$f^*_{ji} = P(X_n = i \text{ for some } n \geq 1 \mid X_0 = j)$$

$$= P\left(\bigcup_{n=1}^{\infty} \{X_n = i\} \,\middle|\, X_0 = j\right)$$

$$\leq \sum_{n=1}^{\infty} \underbrace{P(X_n = i \mid X_0 = j)}_{p^{(n)}_{ji} = 0}$$

$$= 0.$$

$\{A_n : n \geq 1\}$

$B_1 = A_1$

$B_n = A_n \setminus \bigcup_{k=1}^{n-1} A_k$

$\bigcup_{n=1}^{\infty} B_n = \bigcup_{n=1}^{\infty} A_n$

Closed Set :- A subset C of S is called closed if

$$p_{ij} = 0 \text{ for all } i \in C, \; j \notin C.$$

Gambler Ruin prob: $\rightarrow$ $\underbrace{\{0\}, \{N\}}_{\text{Closed}}, \underbrace{\{1, 2, \dots N-1\}}_{\text{not closed}}$.

Gen. Ex: $S = \{0, 1, \dots 7\}$.

$\underbrace{\{0,3\}, \{2,4,7\}}_{\text{closed}}, \underbrace{\{1\}, \{6\}, \{5\}}_{} \rightarrow$ not closed.

The prop. that a rec state can only lead to rec. state implies that the rec. comm. class is closed.

Comm. classes not closed are transient.

Thm :- A chain having a finite no. of states must have at least one rec. state.

Pf:- Sp. not $|S|$ finite.

For any $i \in S$,

$$\sum_{j \in S} p_{ij}^{(n)} = 1$$

On the other hand, if j is transient,

$$\lim_{n \to \infty} p_{ij}^{(n)} = 0$$

Taking limit as $n \to \infty$,
LHS $= 0$ but RHS $= 1$.

$$p_{ij}^{(n)} = \sum_{k=0}^{n} f_{ij}^{(k)} p_{jj}^{(n-k)}$$

$$P_{ij}(s) = F_{ij}(s) P_{jj}(s)$$

$$P_{ij}(s) = F_{ij}(s) P_{jj}(s)$$

$$\Rightarrow \sum_{n \geq 0}^{\infty} p_{ij}^{(n)} = f_{ij}^{*} \sum_{n \geq 0}^{\infty} p_{jj}^{(n)}$$

$\Rightarrow$

For a closed set C,

$$\tilde{P} = ((p_{ij}))_{i,j \in C}.$$

$\tilde{p}_{ij} = p_{ij}$, $i,j \in C$

If C is closed then $p^{(n)}_{ij} = 0$ for all $i \in C$, $j \notin C$.

· $\tilde{P}$ is a tran. matrix.

$$\sum_{j \in C} \tilde{p}_{ij} = 1$$

· $\tilde{P}$ satisfies CK eq^n.

$$p^{(2)}_{ij} = \sum_{k \in S} p_{ik} p_{kj}$$

$$= \sum_{k \in C} p_{ik} p_{kj} + \sum_{k \notin C} p_{ik} p_{kj}$$

$= 0 \qquad 0 \quad \qquad \to 0$

$$\tilde{p}_{ij}^{(n+m)} = \sum_{k \in \tilde{C}} \tilde{p}_{ik}^{(m)} \tilde{p}_{kj}^{(n)}$$

$$\parallel$$

$$p_{ij}^{(m+n)} = \sum_{k \in C} p_{ik}^{(m)} p_{kj}^{(n)}$$

$$= \sum_{k \in C} + \sum_{k \notin C} \quad (= 0)$$

$$= \sum_{k \in C} \tilde{p}_{ik}^{(m)} \tilde{p}_{kj}^{(n)}$$

For $i, j \in \tilde{C}$.

$$\tilde{p}_{ij}^{(n)} = p_{ij}^{(n)}$$

$$p_{ij}^{(2)} = \sum_{k \in S} p_{ik} p_{kj}$$

$$= \sum_{k \in \tilde{C}} p_{ik} p_{kj} + \cancel{\sum_{k \notin C} p_{ik} p_{kj}}$$

$$= \sum_{k \in C} \tilde{p}_{ik} \tilde{p}_{kj} = \tilde{p}_{ij}^{(2)}$$

Consider the Markov on the state sp.
C with transition matrix P.

<u>Thm</u>: A finite closed communicating class is rec

Pf:- Consider the M.C. on the closed comm. class. Since it is finite, there must be at least one rec. state. Since it is a comm. class, all states are of same type. Hence all states are rec.

Thm :- In a finite state sp. chain all rec. states must be pos. rec.

Pf :- Consider a rec. state i and consider comm. class C which contains i.

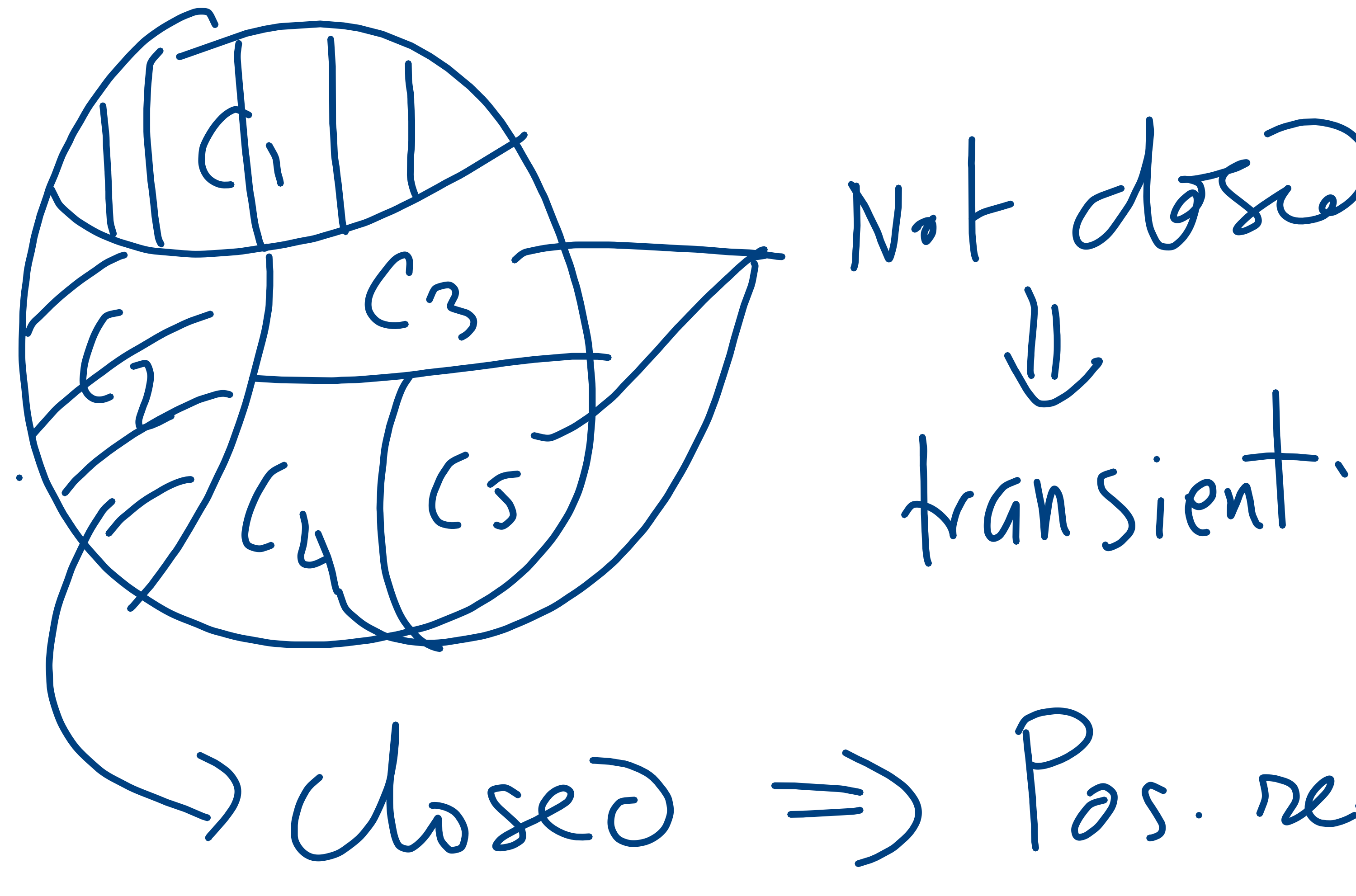
Consider the restricted M.C. on $\mathcal{C}$.
Assume that i is null rec. Then all states in $\mathcal{C}$ are null rec.

$$\sum_{j \in \mathcal{C}} p_{ij}^{(n)} = 1$$

For transient state i, $p_{ij}^{(n)} \to 0$ as $n \to \infty$ for any i & j.
Taking limit as $n \to \infty$, LHS $= 0$, RHS $= 1$.
Hence i is pos. rec.

First Step Analysis:

Gambler's Ruin problem.



$C_1 = \{0\}$, $C_2 = \{N\}$,

$C_3 = \{1, 2, \ldots N-1\}$.

$X_0 = i$, $P(X_n \notin \{0, N\} \; \forall n \geq 1 \mid X_0 = i) = 0$.

$1 \leq i \leq N-1$

$$A = \bigcap_{n=1}^{\infty} \{ X_n \in \bar{C}_3 \} \qquad P(A \mid X_0 = i) = 0.$$

$$\sum_{i \in T} N_i = \infty$$

Let T be any finite set of transient states. Let N_i be the no. of visits to a state i,

$$P(N_i \geq k \mid X_0 = j) = f_{ji}^{*} (f_{ii}^{*})^{k-1}$$

$$\to 0 \text{ as } k \to \infty.$$

$N_i < \infty$ w.p. 1.

$$\Rightarrow \sum_{i \in T} N_i < \infty \text{ w.p. } 1.$$

Since $P\left(\sum_{i\in T} N_i = \infty\right) = 1 - P\left(\sum_{i\in T} N_i < \infty\right)$

$= 1 - 1 = \underline{0}.$